W.H. Croxford High School · Mathematics Department

Mathematics 10-3 Course Outline

2026–27
5 credits

Teacher	Email	Google Classroom	Class website
Mr. Ryan Cathcart Room 2997	rcathcart@rvschools.ab.ca Students and parents welcome	Code given in class Website link, files, announcements	rmcathcart-lab.github.io/math10-3 Lessons, practice, calendar, tests

About the course

Math 10-3 is the first course on the −3 route, built with trades, apprenticeships and the workplace in mind. It is the math you'll use at a job and at home: unit pricing and currency exchange, earning an income, measuring in SI and imperial units, area, volume and capacity, angles and parallel lines, similarity and scale, and right-triangle trigonometry. Every unit ends with a hands-on project.

Math 10-3 → Math 20-3 → Math 30-3

Math 10-3 is worth 5 credits. A high school diploma needs 100 credits, including a 20-level math; Math 20-3 is one.

Success in math is mostly persistence. Everyone in this room can do well in this course by doing the practice, asking questions early and learning from mistakes. Wrong answers are where the learning happens.

A typical class

1. **Arrive ready**: laptop charged, calculator, pencil and paper on your desk.
2. **Warm-up** on the board while attendance is taken.
3. **Lesson**: I teach from the class website. Take notes on paper, or print the notes booklet for the unit.
4. **Practice** on the website: new numbers every time, hints when you're stuck, and your level on each outcome.
5. **Exit slip** almost every day: a short check on today's outcome, written on the website.

Classes run about 88 minutes, bell to bell. The class calendar on the website shows each day's lesson and every test date, and it is always up to date.

The course at a glance

Unit	Topic	Lessons	Outcomes
1	Unit Pricing and Currency Exchange	11	N1
2	Earning an Income	9	N2 · A1
3	Length, Area and Surface Area	12	M1 · M2 · M3 · M4 · A1
4	Volume, Capacity, Mass and Temperature	7	M1 · M2
5	Angles, Parallel Lines and Spatial Puzzles	9	G1 · G5 · G6
6	Similarity and Scale	7	G3
7	Right Triangles and Trigonometry	10	G2 · G4 · A1

65 lessons, a guided project and a unit test in every unit. Dates for your class are on the class calendar.

What you will learn: the 13 outcomes

From Alberta Education's Mathematics 10–12 Program of Studies (2008). These outcomes are what you are assessed on.

Measurement

M1	Demonstrate an understanding of the Système International (SI) by: describing the relationships of the units for length, area, volume, capacity, mass and temperature; applying strategies to convert SI units to imperial units.
M2	Demonstrate an understanding of the imperial system by: describing the relationships of the units for length, area, volume, capacity, mass and temperature; comparing the American and British imperial units for capacity; applying strategies to convert imperial units to SI units.
M3	Solve and verify problems that involve SI and imperial linear measurements, including decimal and fractional measurements.
M4	Solve problems that involve SI and imperial area measurements of regular, composite and irregular 2-D shapes and 3-D objects, including decimal and fractional measurements, and verify the solutions.

Number

N1	Solve problems that involve unit pricing and currency exchange, using proportional reasoning.
N2	Demonstrate an understanding of income, including: wages; salary; contracts; commissions; piecework to calculate gross pay and net pay.

Geometry

G1	Analyze puzzles and games that involve spatial reasoning, using problem-solving strategies.
G2	Demonstrate an understanding of the Pythagorean theorem by: identifying situations that involve right triangles; verifying the formula; applying the formula; solving problems.
G3	Demonstrate an understanding of similarity of convex polygons, including regular and irregular polygons.
G4	Demonstrate an understanding of primary trigonometric ratios (sine, cosine, tangent) by: applying similarity to right triangles; generalizing patterns from similar right triangles; applying the primary trigonometric ratios; solving problems.
G5	Solve problems that involve parallel, perpendicular and transversal lines, and pairs of angles formed between them.
G6	Demonstrate an understanding of angles, including acute, right, obtuse, straight and reflex, by: drawing; replicating and constructing; bisecting; solving problems.

Algebra

A1	Solve problems that require the manipulation and application of formulas related to: perimeter; area; the Pythagorean theorem; primary trigonometric ratios; income.

How you are assessed

Math 10-3 uses **outcomes-based assessment**. Your mark shows which outcomes you can do and how well, not how many points you collected. For each outcome you earn an **indicator**. On a unit test, it reads straight from three boxes of questions:

Beginning box	Progressing box	Mastery box
the core skill	the skill in typical questions	harder, multi-step and new situations

A box is **secure** when you earn ¾ of its points and **half done** at ½:

MAS	Mastery	Beginning, Progressing and Mastery boxes all secure.	85–100%
ADV	Advancing	Beginning and Progressing secure; at least half of Mastery.	75–85%
PRG	Progressing	Beginning and Progressing secure.	65–75%
EMG	Emerging	Beginning secure; at least half of Progressing.	55–65%
BEG	Beginning	Beginning box secure.	45–55%
LIM	Limited	At least half of the Beginning box.	25–45%
NM	Not Meeting	Less than half of the Beginning box.	0–25%
IE	Insufficient Evidence	Not enough work yet to judge, such as a missed test not yet made up.	—

From outcomes to your report card

Part marks and unit grades

Questions can earn part marks: a small slip keeps ⅔ of the point, and a bigger but reasonable mistake keeps ⅓. Your **unit grade** is the average of the unit's outcomes. Exit slips check one or two outcomes from the day's lesson, and they are graded the same way as a unit test: by their Beginning, Progressing and Mastery boxes.

Your final mark

Your report card mark comes from your grades on the 13 outcomes. Everything you do **in class** adds evidence for the outcomes it covers: unit tests, unit projects, exit slips and practice in class. When I make a judgment call on an outcome, I use what I've seen you do in class.

Tests and exit slips

Unit tests and exit slips are written **on the class website**, in class, on your laptop. They pop up full screen when I launch them, and every student gets their own numbers. Tests end at the end of class. **Leaving the test window, switching tabs or leaving full screen submits your work and locks the test**; only I can unlock it.

Once I release the results, you see your answers, the correct answers and a worked solution for each question.

Missed a test?

Make it up at the **Test Centre** in the Learning Commons, Fridays 2:15–3:45. Fill out the missed-assessment form first. If you miss your Test Centre time, you lose that chance.

Practice and due dates

Each lesson's practice is on the website, with a due date on the class calendar. Due dates set the pace, so you're ready for the exit slip and the unit test. Practice isn't marked down for being late, since marks come from work in class, but unfinished practice matters for retests.

Retests: earn another try

A retest is earned with practice, and your **best result counts**, so a retest can only help.

1. **Do the practice.** Finish every required question in the unit without seeing the solution. (Saw one? Redo a new version without it.)
2. **Request a retest.** The button appears on that unit on your home page.
3. **Write it in class.** I open a retest just for you, with new numbers, while the class works.

Want another retest after that? Finish **all the extra practice** in the unit first.

> **Certification cards.** Every outcome has a card beside your name on the website, from Trainee (Limited) up to Master Certified (Mastery). It shows the best level you have reached from practice in class, a project or a unit test.

What to bring every day

- **Laptop or Chromebook**, charged. Croxford is a bring-your-own-device school; ask the office about borrowing a Chromebook for the year.
- **Scientific calculator** like the TI-30XIIS (about $15–20). Not your phone. A few loaners are available in class, and a TI-30XIIS is built into the class website.
- **Pencil, eraser** and **paper or a notebook** for notes (or your printed notes booklet).

Calculators

A calculator is allowed on **everything** in Math 10-3: practice, projects, exit slips and tests. It follows Alberta Education's rules: a calculator is **not allowed** if it gives exact answers such as 3√2 instead of decimals, stores programs or notes, or has anything wireless. Not sure about yours? Bring it in and we'll check it.

Expectations

- **Be on time and ready** to start when the bell goes.
- **Phones off and away** during class (the provincial rule). If you leave the room, your phone stays with me; take the hall pass, one person at a time.
- **Compassionate Cavs:** be kind, encourage each other, and listen when someone is teaching. This is a room where you can be wrong out loud.
- **Own your learning.** Missed a class? The lesson, notes and practice are on the website: catch up before the next class and ask me about anything unclear.

Getting help

Asking early is the biggest predictor of doing well. Ask during work time in class (that's what it's for), use the hints in the practice, or come see me at lunch or before or after school. The only time I won't help is during a test.

Academic integrity

Every assessment must show **your own** thinking. On tests and exit slips, any of these is cheating:

- talking to or communicating with anyone, by any means;
- using notes, devices, websites or apps that weren't allowed;
- using **AI** or photo-solver apps for any part of the work;
- sharing questions or answers with students who haven't written yet.

On practice, copying someone's work costs you the practice. AI and solver apps are fine for *checking* your work, not for producing it.

When academic dishonesty happens, I will talk with you and contact home. It can mean a 0 on that assessment, or an alternate assessment of the outcome at my discretion. Serious or repeated cases are referred to administration.

For parents and guardians

Marks are posted in **PowerSchool** by outcome: an indicator for each outcome a unit test or exit slip has covered, plus each unit grade.

For the day-to-day picture, ask your student to show you their home page on the class website: their level on every outcome, their certification cards, what's due and the next test date.

This course follows Rocky View Schools **AP 360**, Assessment and Communication of Student Learning, and **AP 223**, Student Reassessment (rockyview.ab.ca), and the Croxford student handbook. Questions about this outline? Email rcathcart@rvschools.ab.ca.